

```

# Task 2: Exploratory Data Analysis (EDA)
# Titanic Dataset

# Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset
df = pd.read_csv("Titanic-Dataset.csv.xls")

# -----
# Basic Information
# -----
print("Dataset Shape:", df.shape)

print("\nFirst 5 Rows:")
print(df.head())

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

# -----
# Descriptive Statistics
# -----
print("\nSummary Statistics:")
print(df.describe())

print("\nMedian Values:")
print(df.median(numeric_only=True))

# -----
# Correlation Matrix
# -----
numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,6))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Matrix")
plt.show()

# -----
# Histograms
# -----
numeric_df.hist(figsize=(12,10), bins=20)
plt.suptitle("Histograms of Numerical Features")
plt.show()

# -----
# Boxplots
# -----
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.boxplot(y=df["Age"])
plt.title("Boxplot - Age")

plt.subplot(1,2,2)
sns.boxplot(y=df["Fare"])
plt.title("Boxplot - Fare")

plt.tight_layout()
plt.show()

# -----
# Pairplot
# -----
sns.pairplot(
    df[["Survived", "Pclass", "Age", "Fare"]].dropna(),
    hue="Survived"
)
plt.show()

```

```

# -----
# Survival Distribution
# -----
plt.figure(figsize=(6,4))
sns.countplot(x="Survived", data=df)
plt.title("Survival Distribution")
plt.show()

# -----
# Passenger Class Distribution
# -----
plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", data=df)
plt.title("Passenger Class Distribution")
plt.show()

# -----
# Gender Distribution
# -----
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", data=df)
plt.title("Gender Distribution")
plt.show()

# -----
# Survival by Gender
# -----
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.show()

# -----
# Survival by Passenger Class
# -----
plt.figure(figsize=(6,4))
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.show()

# -----
# Fare Distribution
# -----
plt.figure(figsize=(8,5))
sns.histplot(df["Fare"], bins=30, kde=True)
plt.title("Fare Distribution")
plt.show()

# -----
# Age Distribution
# -----
plt.figure(figsize=(8,5))
sns.histplot(df["Age"], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()

# -----
# Missing Values Visualization
# -----
plt.figure(figsize=(8,5))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Values Heatmap")
plt.show()

# -----
# Skewness Detection
# -----
print("\nSkewness of Numerical Features:")
print(numeric_df.skew())

# -----
# EDA Insights
# -----
print("\nEDA Insights:")
print("1. Female passengers had a higher survival rate than males.")
print("2. First-class passengers survived more often than lower classes.")
print("3. Fare contains several outliers.")
print("4. Age has some missing values and is slightly right-skewed.")
print("5. Cabin column contains many missing values.")
print("6. Fare and Passenger Class show relationship with survival.")

```


Dataset Shape: (891, 12)

First 5 Rows:

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null	Count	Dtype
0	PassengerId	891 non-null		int64
1	Survived	891 non-null		int64
2	Pclass	891 non-null		int64
3	Name	891 non-null		object
4	Sex	891 non-null		object
5	Age	714 non-null		float64
6	SibSp	891 non-null		int64
7	Parch	891 non-null		int64
8	Ticket	891 non-null		object
9	Fare	891 non-null		float64
10	Cabin	204 non-null		object
11	Embarked	889 non-null		object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

Missing Values:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Summary Statistics:

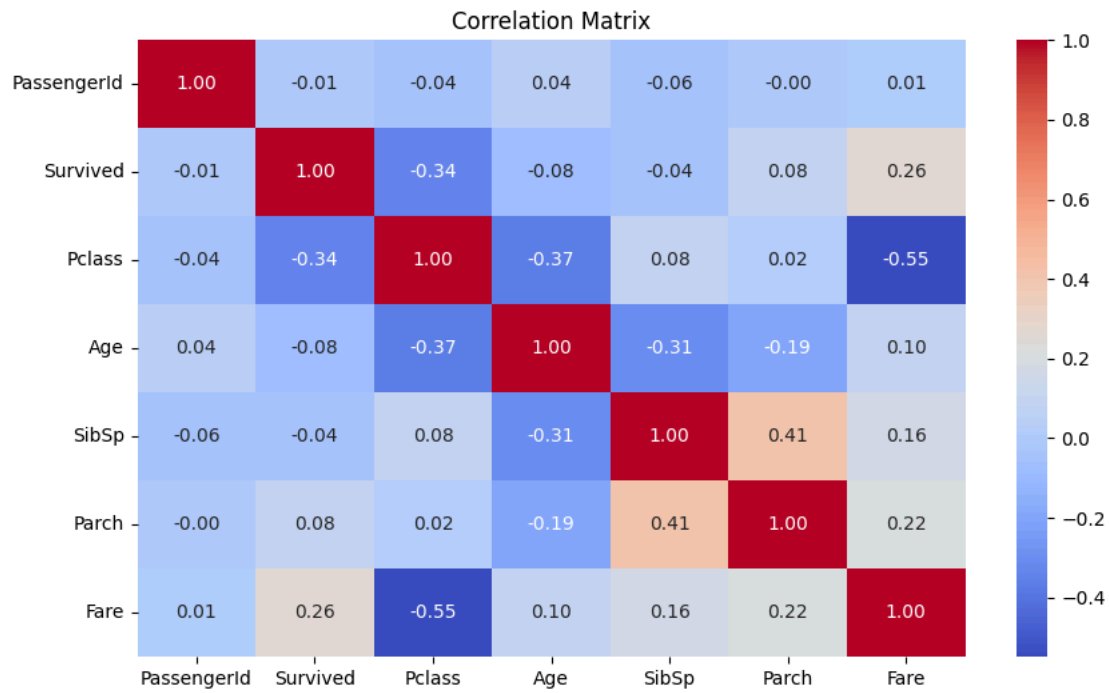
	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Median Values:

PassengerId	446.0000
Survived	0.0000
Pclass	3.0000
Age	28.0000
SibSp	0.0000
Parch	0.0000
Fare	14.4542

dtype: float64



Histograms of Numerical Features

